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11.3.4 Turgor Pressure and Wall Pressure

When a plant cell is placed in pure water (hypotonic solution) the diffusion of water into the cell takes place by endosmosis. It creates a positive hydrostatic pressure on the rigid cell wall by the cell membrane. Henceforth the pressure exerted by the cell membrane towards the cell wall is **Turgor Pressure (TP).**

The cell wall reacts to this turgor pressure with **equal and opposite force**, and the counter-pressure exerted by the cell wall towards cell membrane is **wall pressure (WP).**

Turgor pressure and wall pressure make the cell fully turgid.

TP + WP = Turgid.

Activity

Find the role of turgor pressure in sudden closing of leaves when we touch the 'touch me not' plant.

11.3.5 Diffusion Pressure Deficit (DPD) or Suction Pressure (SP)

Pure solvent (hypotonic) has higher diffusion pressure. Addition of solute in pure solvent lowers its diffusion pressure. The difference between the diffusion pressure of the solution and its solvent at a particular temperature and atmospheric pressure is called as **Diffusion Pressure Deficit (DPD)** termed by **Meyer (1938)**. DPD is increased by the addition of solute into a solvent system. Increased DPD favours endosmosis or it sucks the water from hypotonic solution; hence **Renner** (1935) called it as **Suction pressure**. It is equal to the difference of osmotic pressure and turgor pressure of a cell. The following three situations are seen in plants:

- **DPD in normal cell:** DPD = OP – TP.
- **DPD in fully turgid cell**: Osmotic pressure is always equal to turgor pressure in a fully turgid cell.
- OP = TP or OP-TP =0. Hence DPD of fully turgid cell is zero.
- **DPD in flaccid cell**: If the cell is in flaccid condition there is no turgor pressure or TP=0. Hence DPD = OP.

11.3.6 Osmosis

Osmosis (Latin: *Osmos*-impulse, urge) is a **special type of diffusion**. It represents the movement of **water or solvent molecules** through a selectively permeable membrane **from the place of its higher concentration (high water potential) to the place of its lower concentration (low water potential).**

Types of Solutions based on concentration

i. ***Hypertonic*** (*Hyper* = High; *tonic* = solute): This is a strong solution (low solvent/ high solute / low **Ψ**) which **attracts solvent** from other solutions.

ii. ***Hypotonic*** (*Hypo* = low; *tonic* = solute): This is a weak solution (high solvent /low or zero solute / high **Ψ**) and it **diffuses water** out to other solutions (Figure 11.7).

iii. ***Isotonic*** (*Iso* = identical; *tonic* = soute): It refers to two solutions having same concentration. In this condition the net movement of water molecule will be zero.

The term hyper, hypo and isotonic are **relative terms** which can be used only